



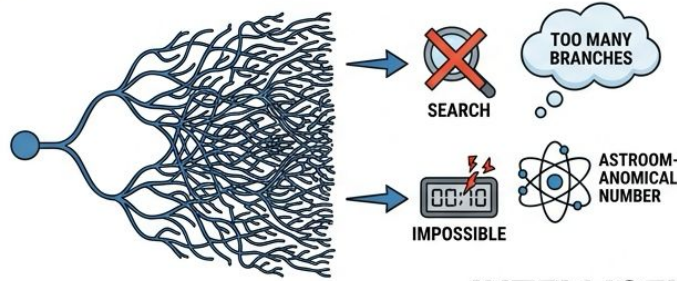
U.S. General Services  
Administration

# Chapter 5.5: Monte Carlo Tree Search Fundamentals

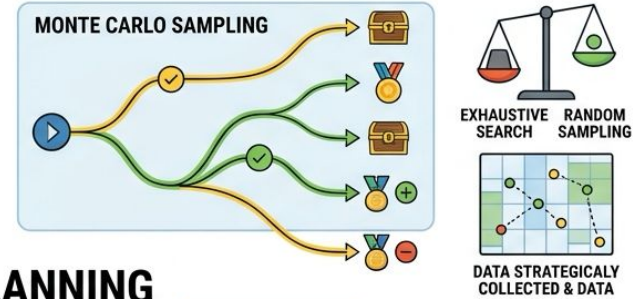
US General Service Administration  
AI Community of Practice - March 2026

# Issues with Vast Action Spaces

## 1 EXPONENTIAL ACTION SPACES DEFEAT EXHAUSTIVE SEARCH

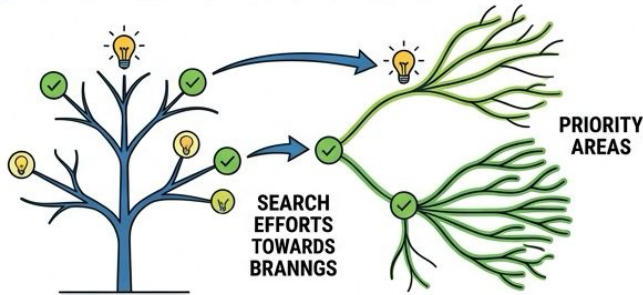


## 2 RANDOM SAMPLING ESTIMATES WHAT BRUTE FORCE CANNOT COMPUTE

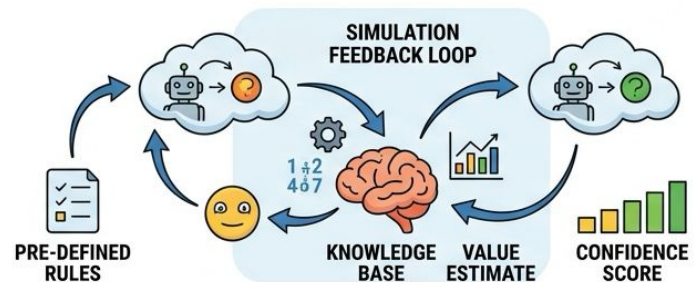


## INTELLIGENT PLANNING IN VAST ACTION SPACES

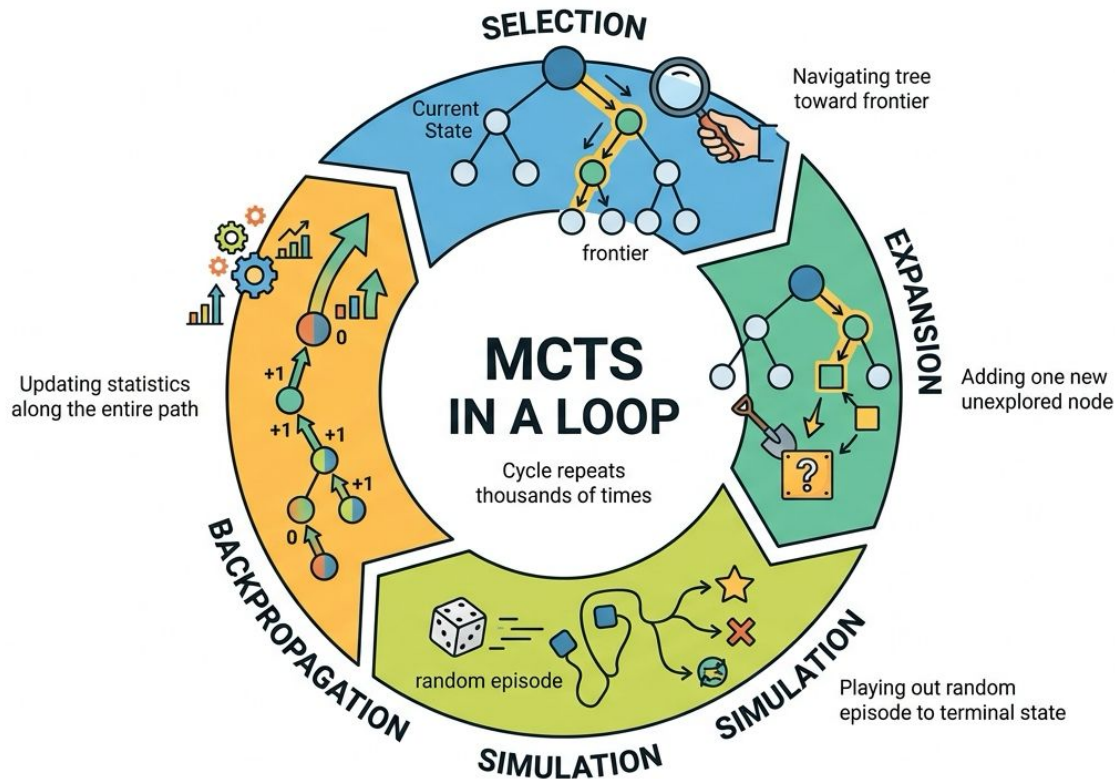
## 3 TREE STRUCTURE FOCUSES EFFORT ON PROMISING BRANCHES



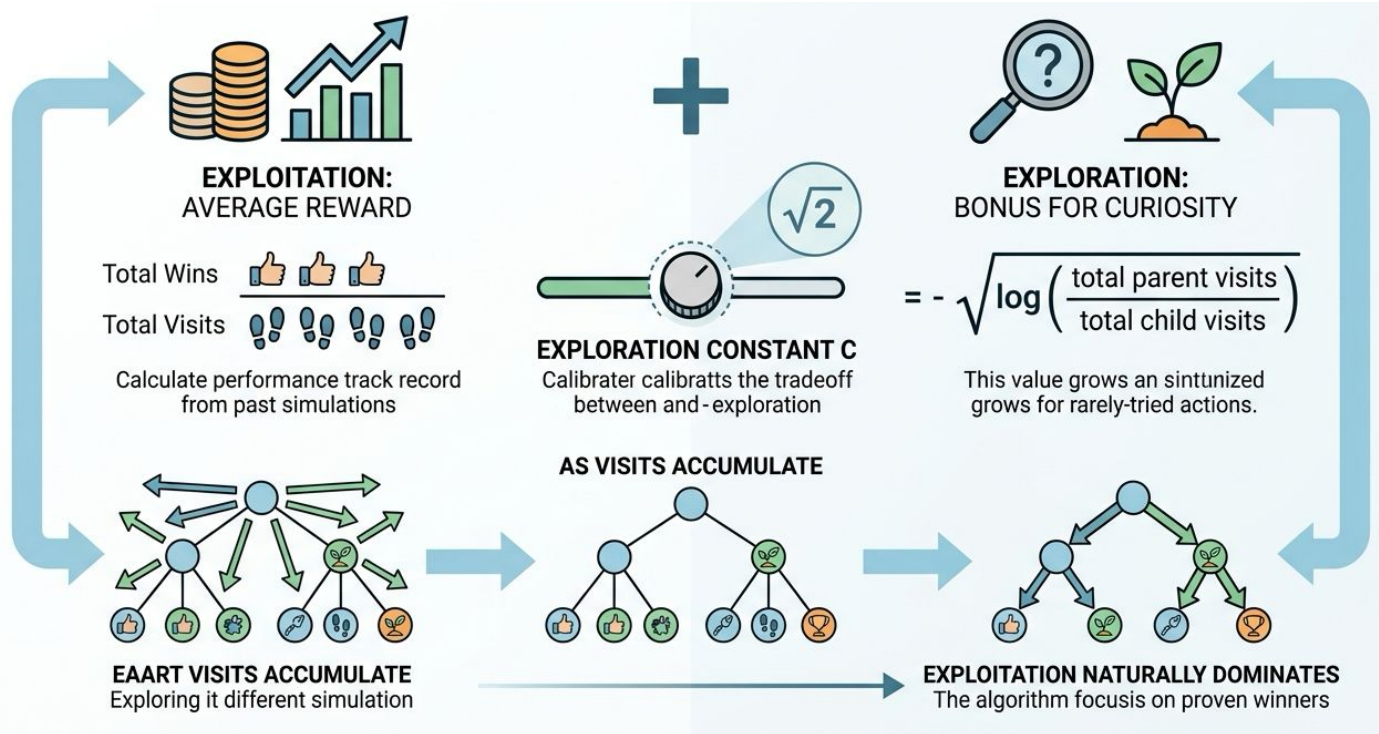
## MCTS LEARNS VALUE FROM SIMULATED EXPERIENCE, NOT HEURISTICS



# MCTS in a Loop

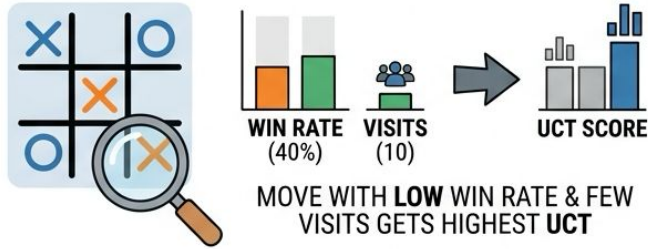


# Ballancing Curiosity and Confidence

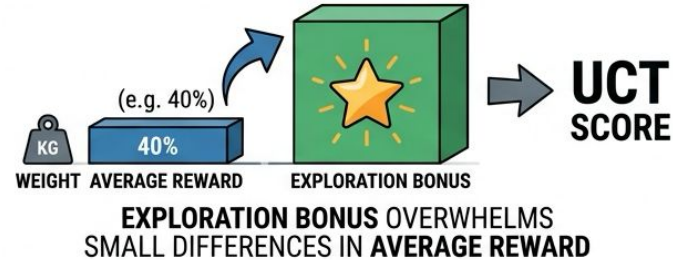


# A Worked Example -- Tic-Tac-Toe UCT Scores

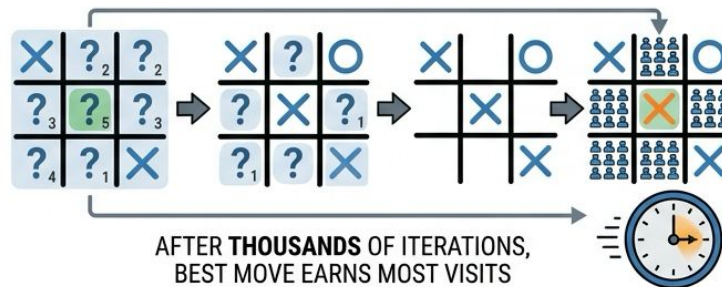
## 1. HIGH EXPLORATION ON UNDER-VISITED MOVE



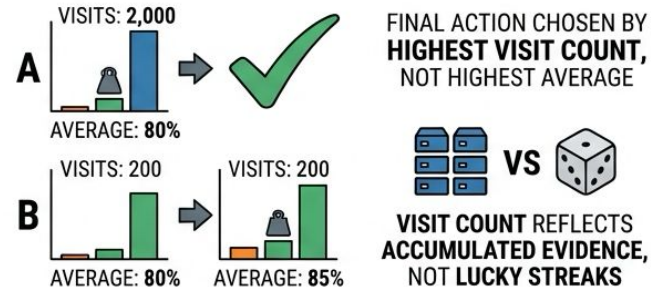
## 2. EXPLORATION BONUS OVERWHELMS AVERAGE REWARD



## 3. ALGORITHM ACCUMULATES EVIDENCE OVER TIME



## 4. FINAL DECISION BY VISITS, NOT AVERAGE





# Structural Elements of Search Trees

## WHAT EACH NODE ACTUALLY STORES

### TREE NODE

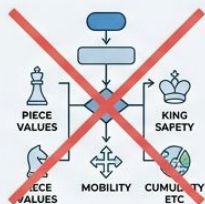


VISIT  
COUNT

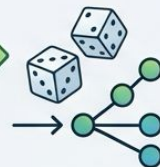


CUMULATIVE  
REWARD

## NO HAND-CRAFTED EVALUATIONS NEEDED

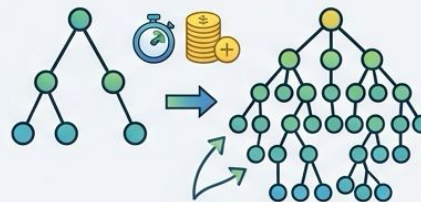


TRADITIONAL  
EVALUATION



LEARNED VALUE  
FROM SIMULATIONS

## INCREMENTAL TREE GROWTH

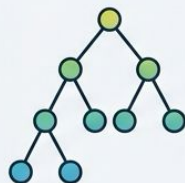


FEW  
SIMULATIONS

DENSE BRANCHES  
= PROMISING PATHS

MANY  
SIMULATIONS

## TREE REUSE

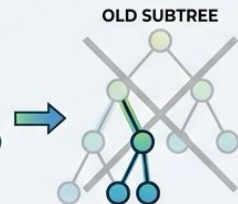


ORIGINAL TREE

AGENT'S  
ACTION  
TAKEN

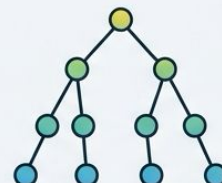


NEW STATE

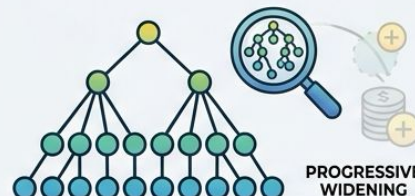


OLD SUBTREE  
RETAINED  
KNOWLEDGE MAP  
PRESERVES PREVIOUS LEARNING

## IMPACT OF BRANCHING FACTOR



LOW BRANCHING FACTOR  
FASTER CONVERGENCE

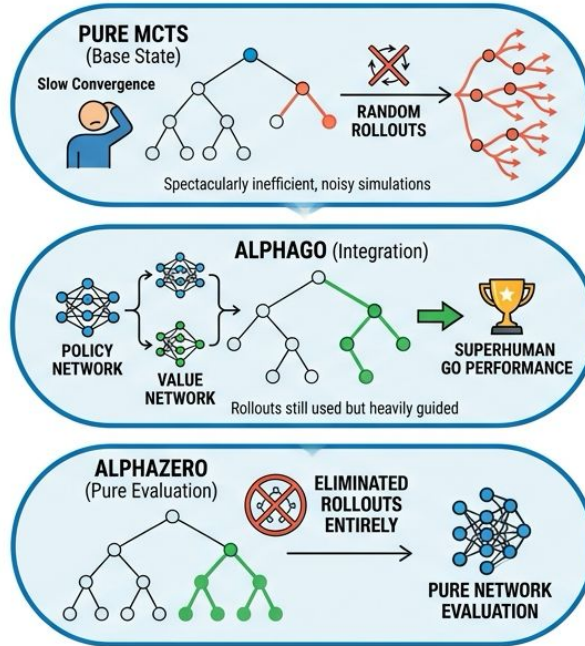


HIGH BRANCHING FACTOR  
REQUIRES MORE SIMULATIONS

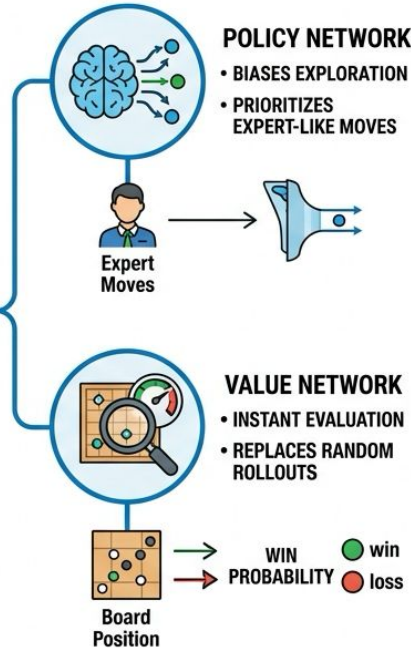
PROGRESSIVE  
WIDENING

# Neural Network Enhancement

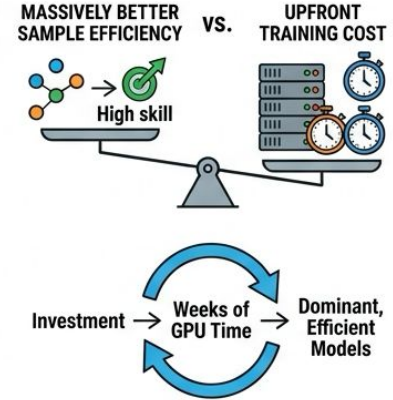
## EVOLUTION OF INTELLIGENT SEARCH



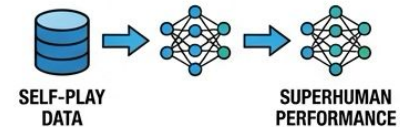
## TWO KEY NEURAL NETWORKS



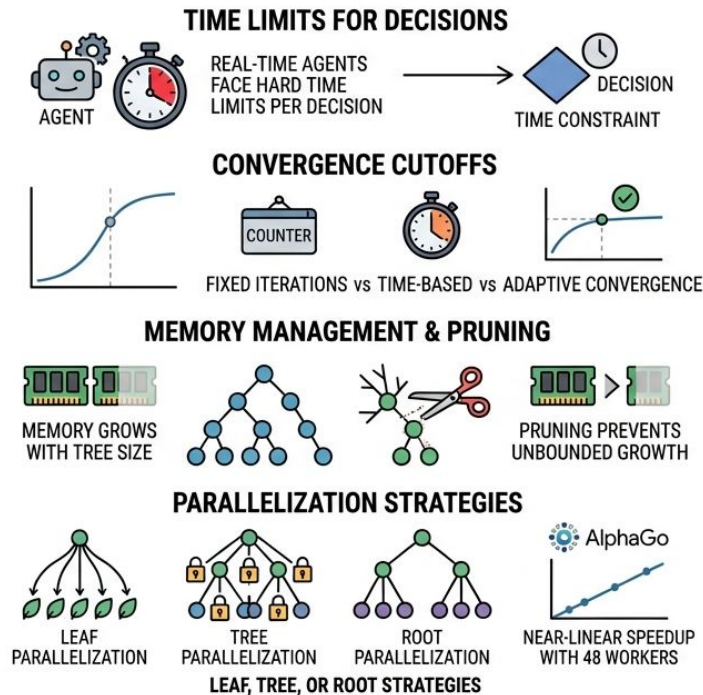
## SYSTEM TRADEOFF



## ACHIEVING DOMINANCE



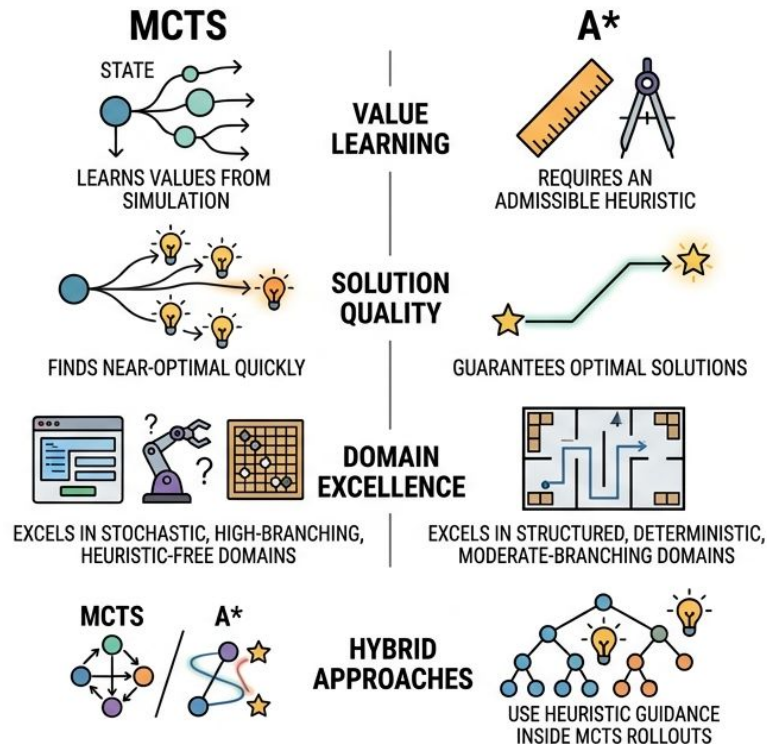
# Time, Memory, and Parallelism



- Real-time agents face hard time limits per decision
- Fixed iteration count vs. time-based vs. adaptive convergence cutoffs
- Memory grows with tree size -- pruning prevents unbounded growth
- Parallelization: leaf, tree, or root strategies
- AlphaGo achieved near-linear speedup with 48 workers



# MCTS vs. A\*



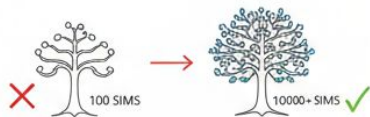
- A\* requires an admissible heuristic; MCTS learns values from simulation
- A\* guarantees optimal solutions; MCTS finds near-optimal quickly
- A\* excels in structured, deterministic, moderate-branching domains
- MCTS excels in stochastic, high-branching, heuristic-free domains
- Hybrid approaches use heuristic guidance inside MCTS rollouts

# Implementation Pitfalls & Rollout Policies

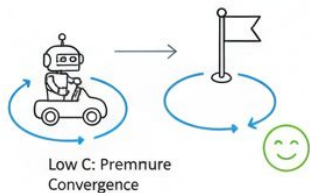


## IMPLEMENTATION PITFFALS

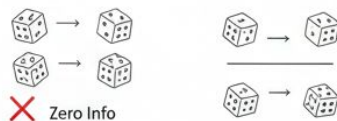
### Insufficient Simulation Budget



### Misaligned Reward Functions



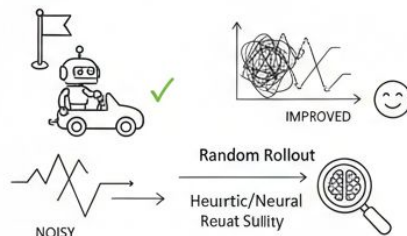
### Deterministic RNG Seeds



### Poorly Tuned \*C\*



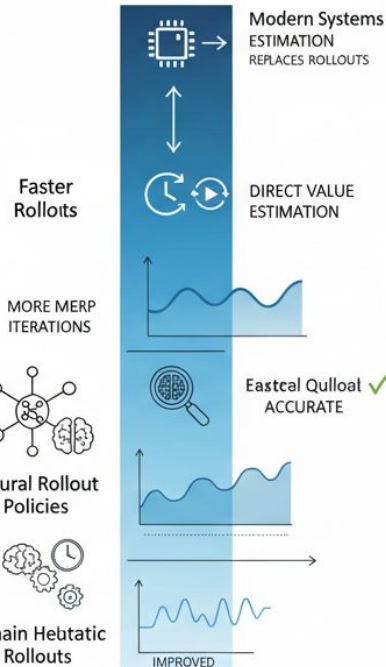
### Misaligned Reward Functions



### Random Rollout Noise

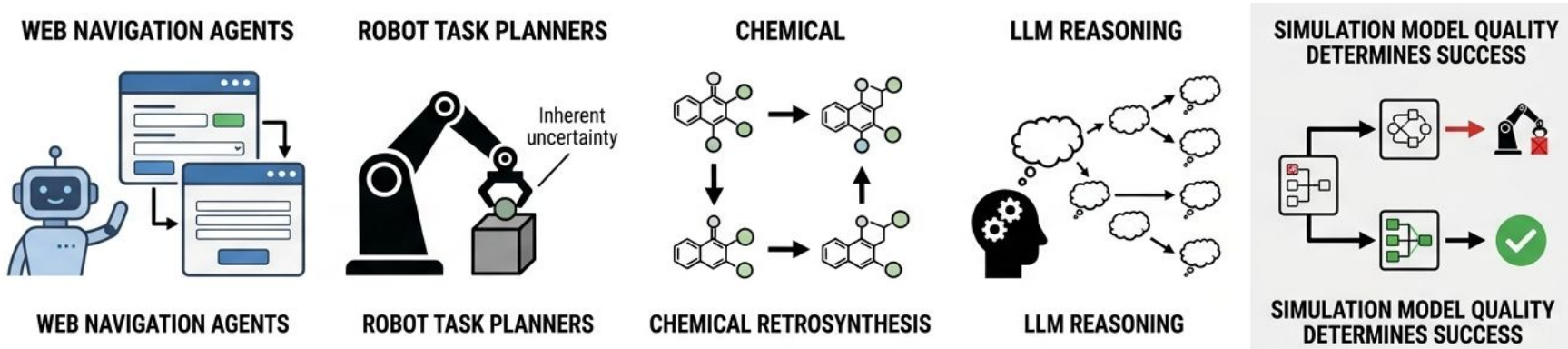


## ROLLOUT POLICIES



From Random to Informed

# MCTS for Agentic AI -- Beyond Board Games



- Web navigation agents explore page interaction sequences
- Robot task planners handle uncertain grasp and manipulation
- Chemical retrosynthesis searches reaction pathways
- LLM reasoning uses MCTS to explore chain-of-thought branches
- Simulation model quality determines real-world MCTS success

# Key Takeaways

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- MCTS explores by sampling, not exhaustive enumeration
  - UCT automatically balances exploration and exploitation
  - Neural guidance transforms MCTS from competent to superhuman
  - Budget, memory, and parallelism are production-critical concerns
  - Next chapter: A\* pathfinding and replanning for structured domains
- 